PRELIMINARY — NOT APPROVED FOR FABRICATION OR ENERGIZATION

PROJECT BUTTON HR-V0 ELECTRICAL V3 CONNECTED CANDIDATE

Separate RESET and ARM, two PNOZ stages, dual watchdog contacts, external adapters, redundant actuator interruption.

01 External listed sources and DC boundaries

File: 01_external_sources.kicad_sch

02 Dual-channel E-stop and RESET eligibility

File: 02_estop_eligibility.kicad_sch

03 Distinct ARM and watchdog eligibility

File: 03_arm_watchdog_eligibility.kicad_sch

04 Contactor coils, mirror contacts and EDM

File: 04_contactor_edm.kicad_sch

05 Redundant actuator-power interruption

File: 05_actuator_interruption.kicad_sch

06 Protected actuator branches and VDD-isolating injection

File: 06_branches_and_injection.kicad_sch

07 Independent watchdog power, controller and drivers

File: 07_watchdog_control.kicad_sch

08 Compute, debug and control terminals

File: 08_compute_and_control_terminals.kicad_sch

09 U2D2, actuator ports and bonding boundary

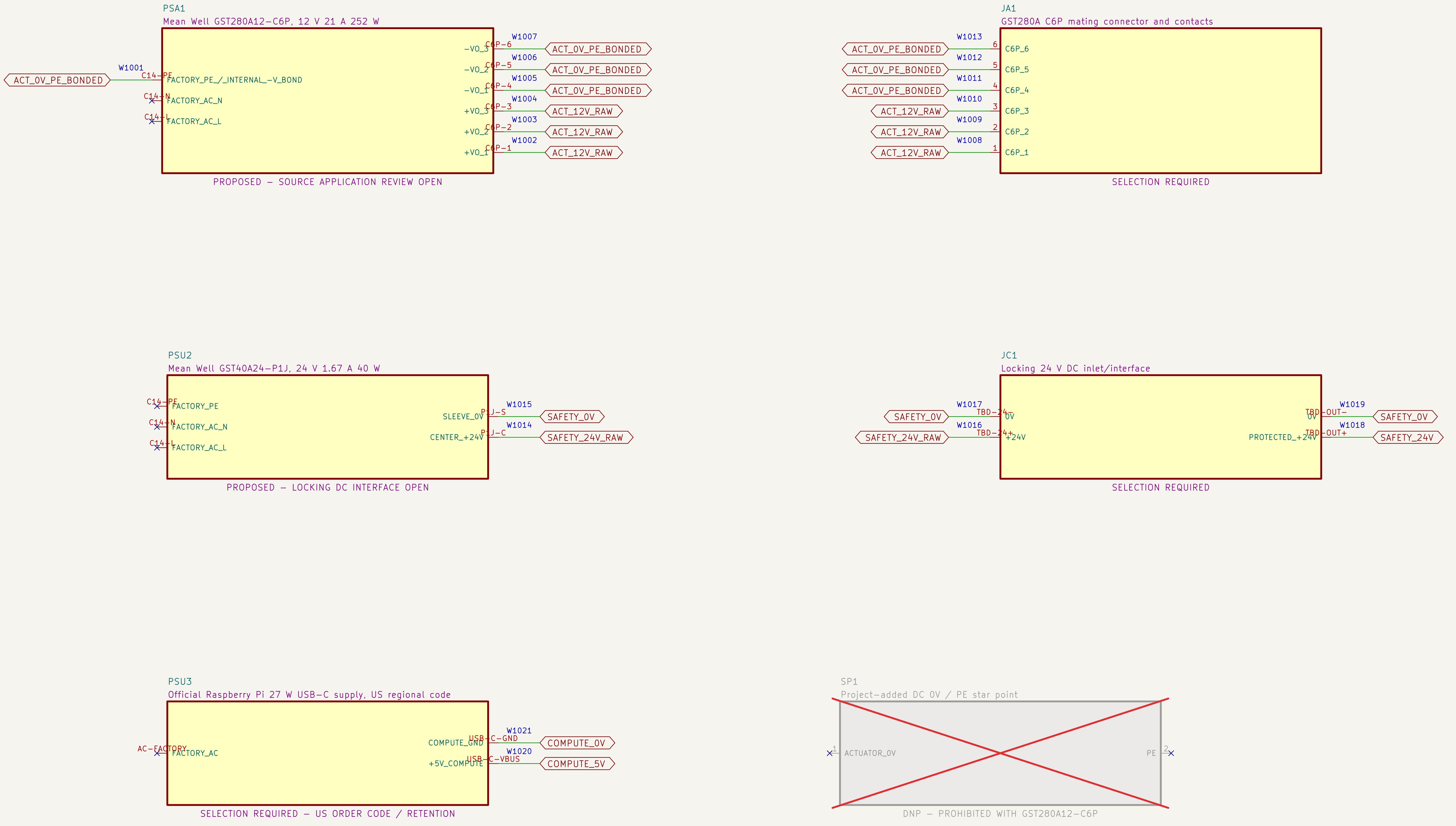
File: 09_actuator_interfaces.kicad_sch

V2.1 PRESERVED; V3 IS A CONNECTED CANDIDATE
PRELIMINARY — NOT APPROVED FOR FABRICATION OR ENERGIZATION
Project Button
Sheet: /
File: project-button-v3.kicad_sch
Title: Project Button HR-V0 Electrical V3 index

Size: A3	Date: 2026-08-06	**Rev: V3-P0.2**
KiCad E.D.A. 10.0.5		Id: 1/10

01 External listed sources and DC boundaries

Factory–sealed adapters eliminate project–built mains wiring.



NOTE 1: All AC conductors shown are factory boundaries, not project–built wiring.

NOTE 2: Site cords, receptacles, GFCI/code basis and source application review remain open.

CONNECTED DESIGN CANDIDATE – QUALIFIED REVIEW REQUIRED
PRELIMINARY – NOT APPROVED FOR FABRICATION OR ENERGIZATION
Project Button
Sheet: /01 External listed sources and DC boundaries/
File: 01_external_sources.kicad_sch

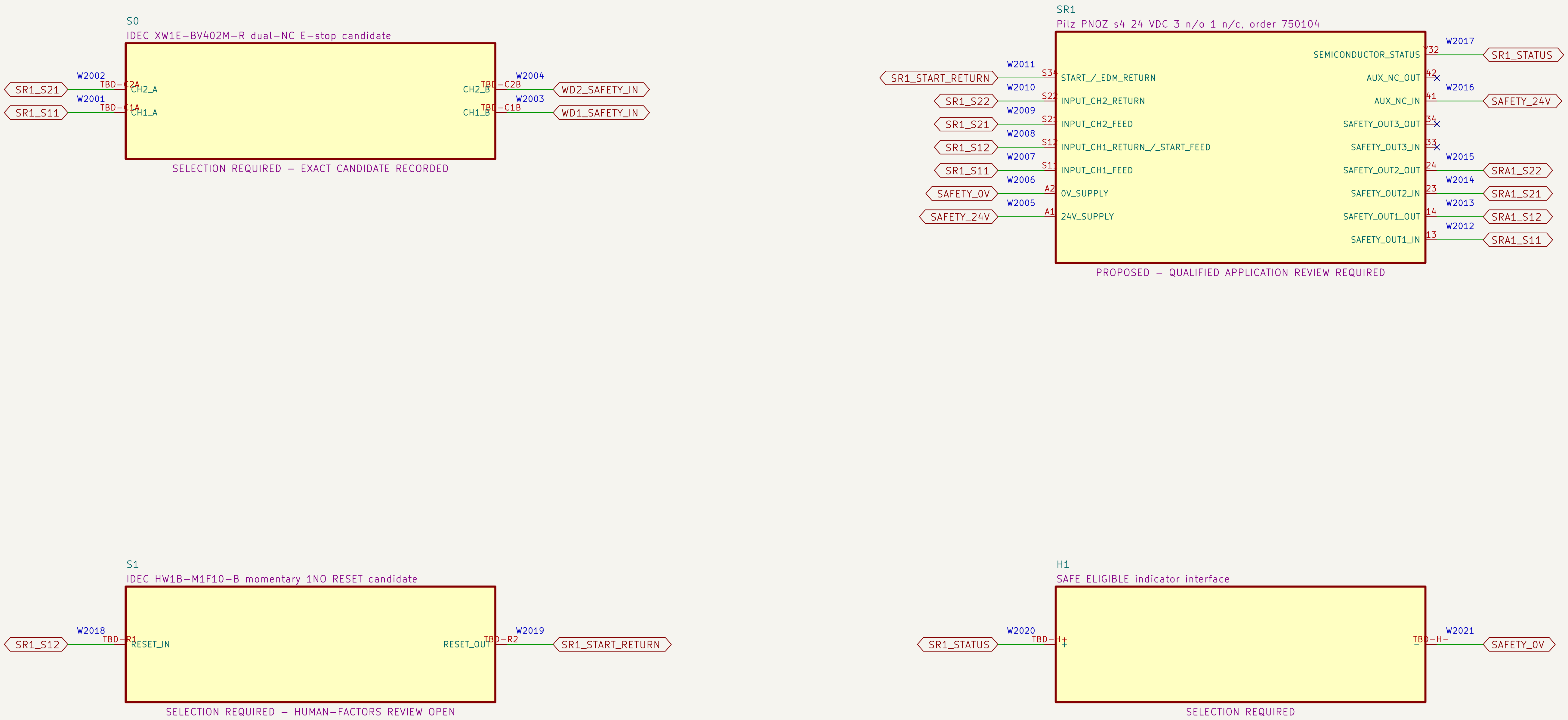
Title: PB HR–V0 Electrical V3 – 01

Size: A3	Date: 2026–08–06	**Rev: V3–P0.2**
KiCad E.D.A. 10.0.5		Id: 2/10

PRELIMINARY — NOT APPROVED FOR FABRICATION OR ENERGIZATION

02 Dual-channel E-stop and RESET eligibility

Each SR1 input return contains one E-stop NC and one watchdog NO contact; RESET cannot energize K1/K2.



NOTE 1: Heartbeat loss opens both SR1 input returns; recovery alone cannot restore the monitored RESET stage.

NOTE 2: RESET release may make SR1 eligible, but SRA1 and K1/K2 remain de-energized until a later ARM.

CONNECTED DESIGN CANDIDATE — QUALIFIED REVIEW REQUIRED
PRELIMINARY — NOT APPROVED FOR FABRICATION OR ENERGIZATION

Project Button

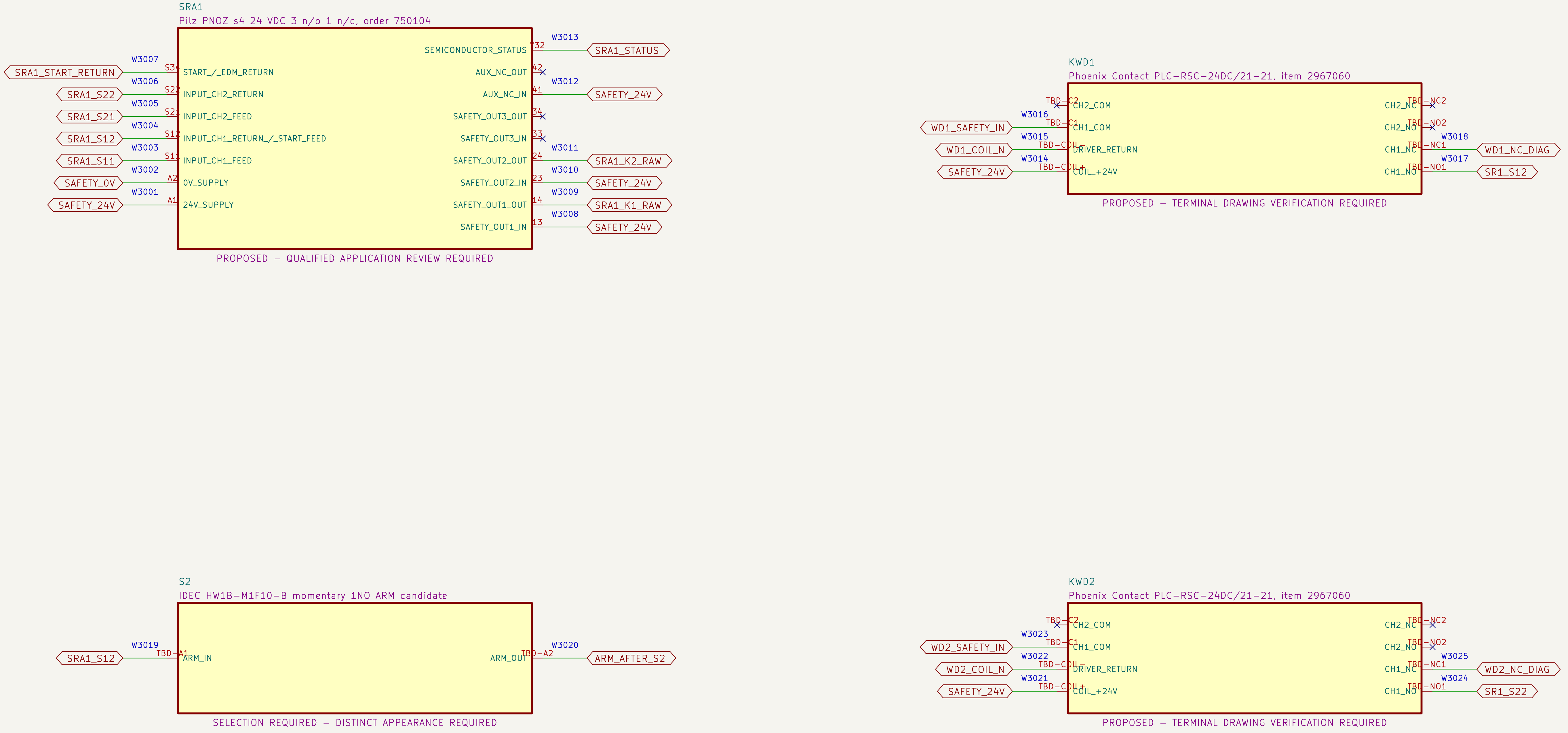
Sheet: /02 Dual-channel E-stop and RESET eligibility/
File: 02_estop_eligibility.kicad_sch

Title: PB HR-V0 Electrical V3 — 02

Size: A3	Date: 2026-08-06	**Rev: V3-P0.2**
KiCad E.D.A. 10.0.5		Id: 3/10

03 Distinct ARM and watchdog eligibility

SRA1 requires SR1 eligibility, two watchdog channels, EDM proof and a new ARM action.



NOTE 1: Required after E-stop/watchdog dropout: cause healthy -> RESET press/release ->
SAFE_READY -> distinct ARM press/release.

NOTE 2: KWD contacts are in the SR1 returns; SRA1 receives the two SR1 safety outputs
directly.

CONNECTED DESIGN CANDIDATE — QUALIFIED REVIEW REQUIRED
PRELIMINARY — NOT APPROVED FOR FABRICATION OR ENERGIZATION

Project Button

Sheet: /03 Distinct ARM and watchdog eligibility/
File: 03_arm_watchdog_eligibility.kicad_sch

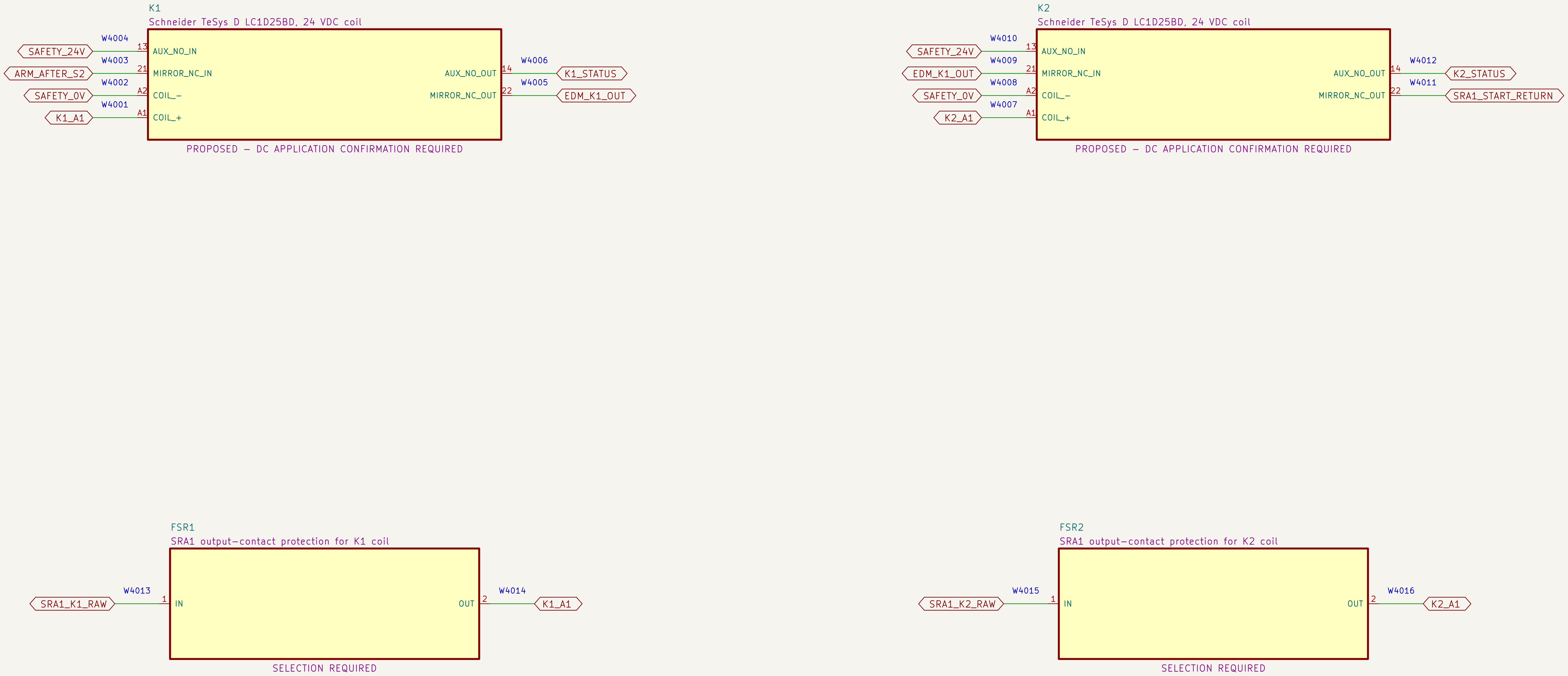
Title: PB HR-V0 Electrical V3 — 03

Size: A3	Date: 2026-08-06	**Rev: V3-P0.2**
KiCad E.D.A. 10.0.5		Id: 4/10

PRELIMINARY — NOT APPROVED FOR FABRICATION OR ENERGIZATION

04 Contactor coils, mirror contacts and EDM

K1 and K2 are distinct final elements; their mirror contacts form the monitored restart return.



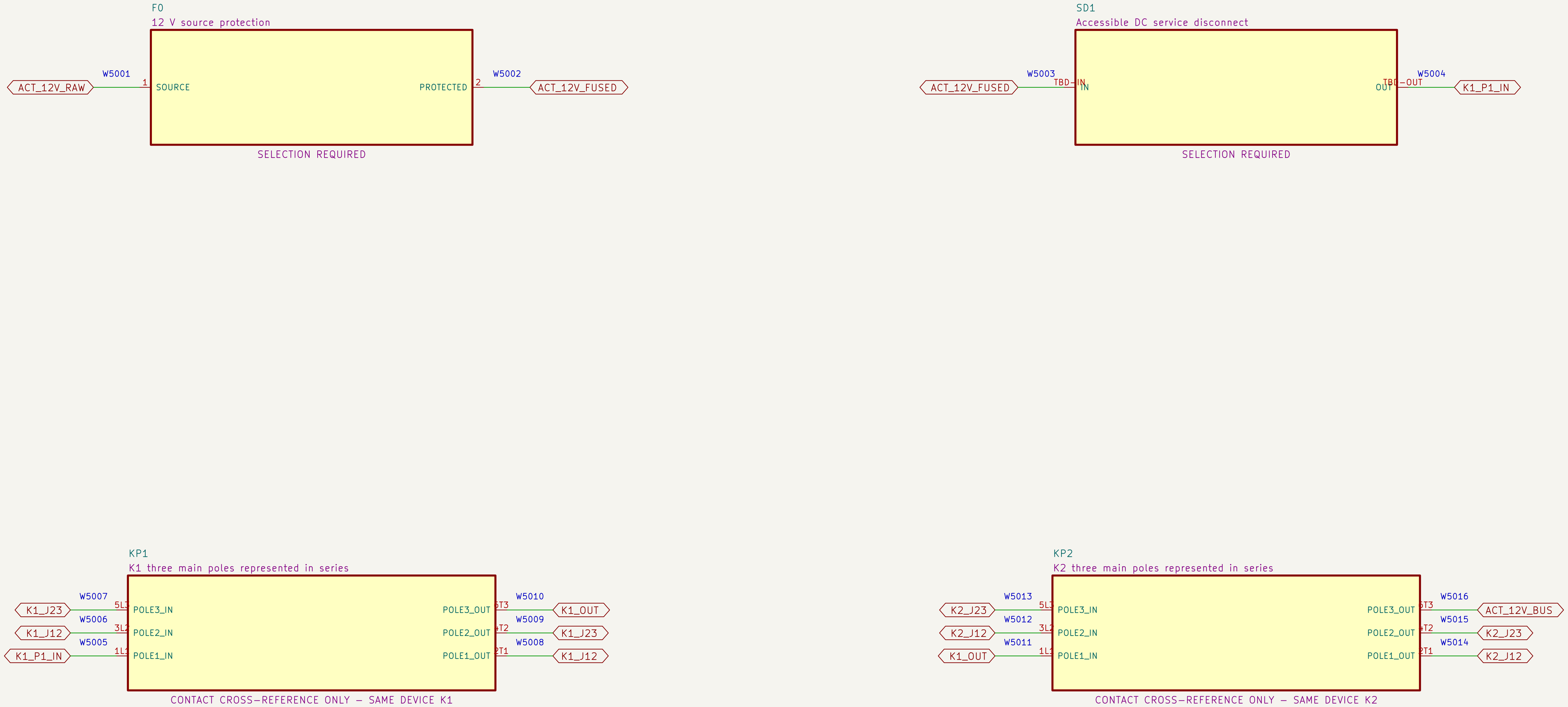
NOTE 1: SRA1 outputs are separately protected before K1 and K2 coils; exact protection remains selection required.

NOTE 2: Loaded DC interruption, suppression behavior, mirror−contact use and coordination require qualified review.

CONNECTED DESIGN CANDIDATE — QUALIFIED REVIEW REQUIRED
PRELIMINARY — NOT APPROVED FOR FABRICATION OR ENERGIZATION
Project Button
Sheet: /04 Contactor coils, mirror contacts and EDM/
File: 04_contactor_edm.kicad_sch
Title: PB HR−V0 Electrical V3 — 04
Size: A3 | Date: 2026−08−06 | **Rev: V3−P0.2**
KiCad E.D.A. 10.0.5 | Id: 5/10

05 Redundant actuator–power interruption

Source protection, service disconnect and all three poles of K1 then K2 are represented in series.



NOTE 1: Series jumpers: K1 2T1–>3L2, 4T2–>5L3, 6T3–>K2 1L1; repeat through K2 to ACT_12V_BUS.

NOTE 2: No fuse, disconnect, conductor, connector or contactor application is released without fault-current evidence.

CONNECTED DESIGN CANDIDATE – QUALIFIED REVIEW REQUIRED
PRELIMINARY – NOT APPROVED FOR FABRICATION OR ENERGIZATION

Project Button

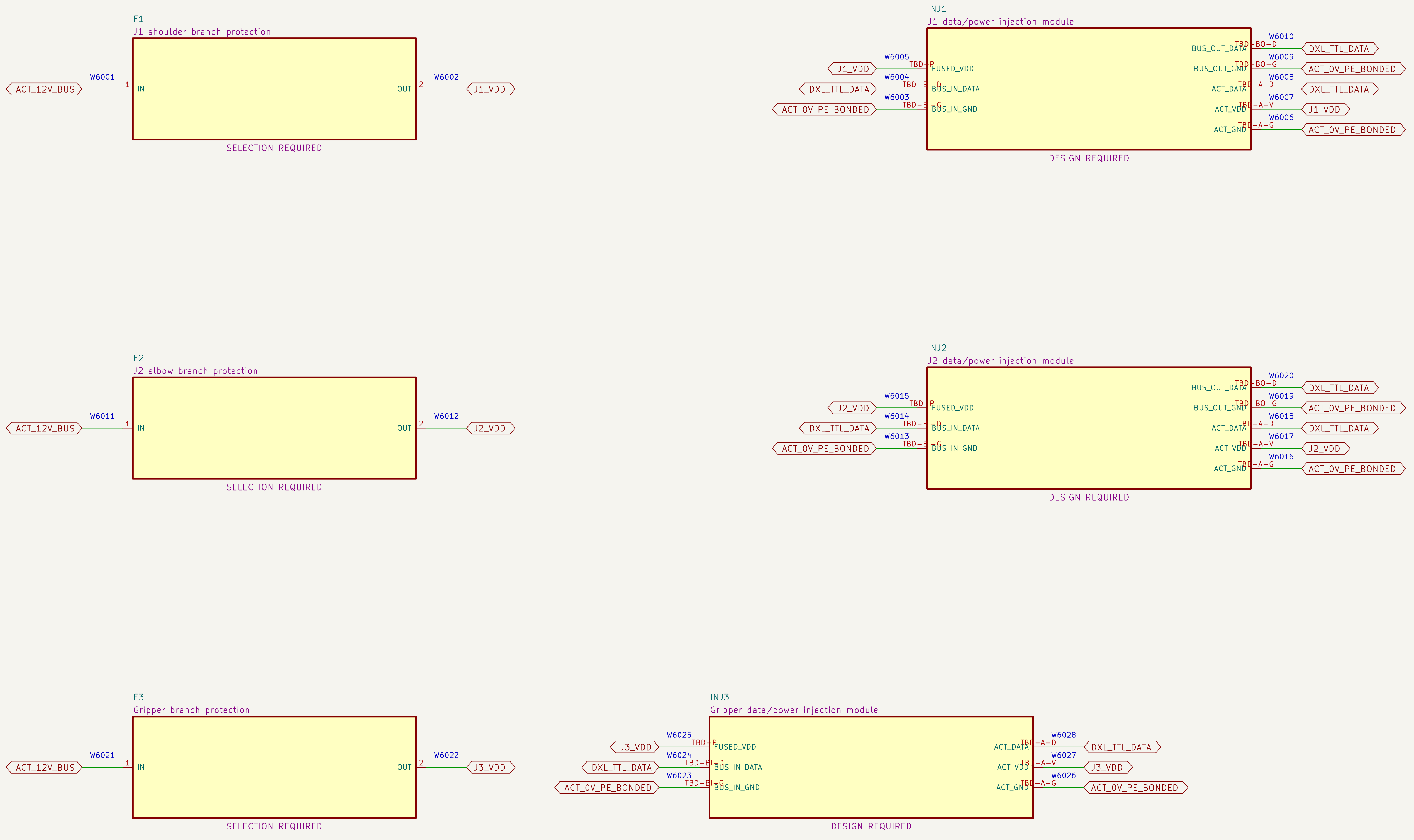
Sheet: /05 Redundant actuator–power interruption/
File: 05_actuator_interruption.kicad_sch

Title: PB HR-V0 Electrical V3 – 05

Size: A3	Date: 2026-08-06	Rev: V3-P0.2
KiCad E.D.A. 10.0.5		Id: 6/10

06 Protected actuator branches and VDD–isolating injection

Each actuator has a separate protected VDD branch; inter–actuator data cables must omit VDD.



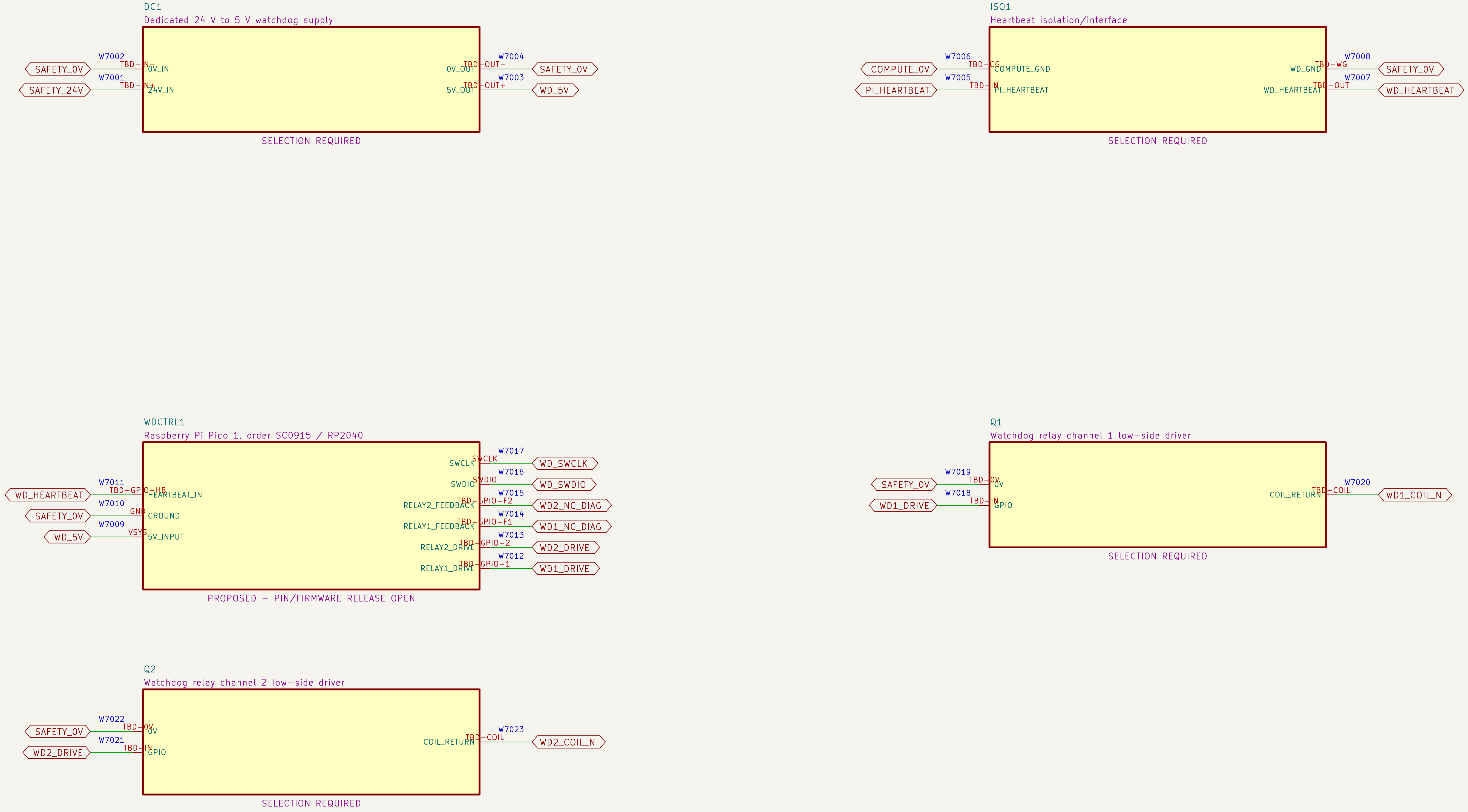
NOTE 1: Standard fully populated ROBOTIS TTL daisy–chain cables are prohibited because they would parallel branch VDD.

NOTE 2: Every injection module requires released source, continuity, isolation, pull and no–backfeed tests.

CONNECTED DESIGN CANDIDATE — QUALIFIED REVIEW REQUIRED
PRELIMINARY — NOT APPROVED FOR FABRICATION OR ENERGIZATION
Project Button
Sheet: /06 Protected actuator branches and VDD–isolating injection/
File: 06_branches_and_injection.kicad_sch
Title: PB HR–V0 Electrical V3 — 06
Size: A3 Date: 2026–08–06 **Rev: V3–P0.2**
KiCad E.D.A. 10.0.5 Id: 7/10

07 Independent watchdog power, controller and drivers

The watchdog forces restart but receives no safety-integrity credit by assertion.



NOTE 1: Power-up, brownout, clock failure, stuck GPIO, held heartbeat and firmware-corruption tests are mandatory.

NOTE 2: A qualified review must decide whether watchdog loss is credited or diagnostic only.

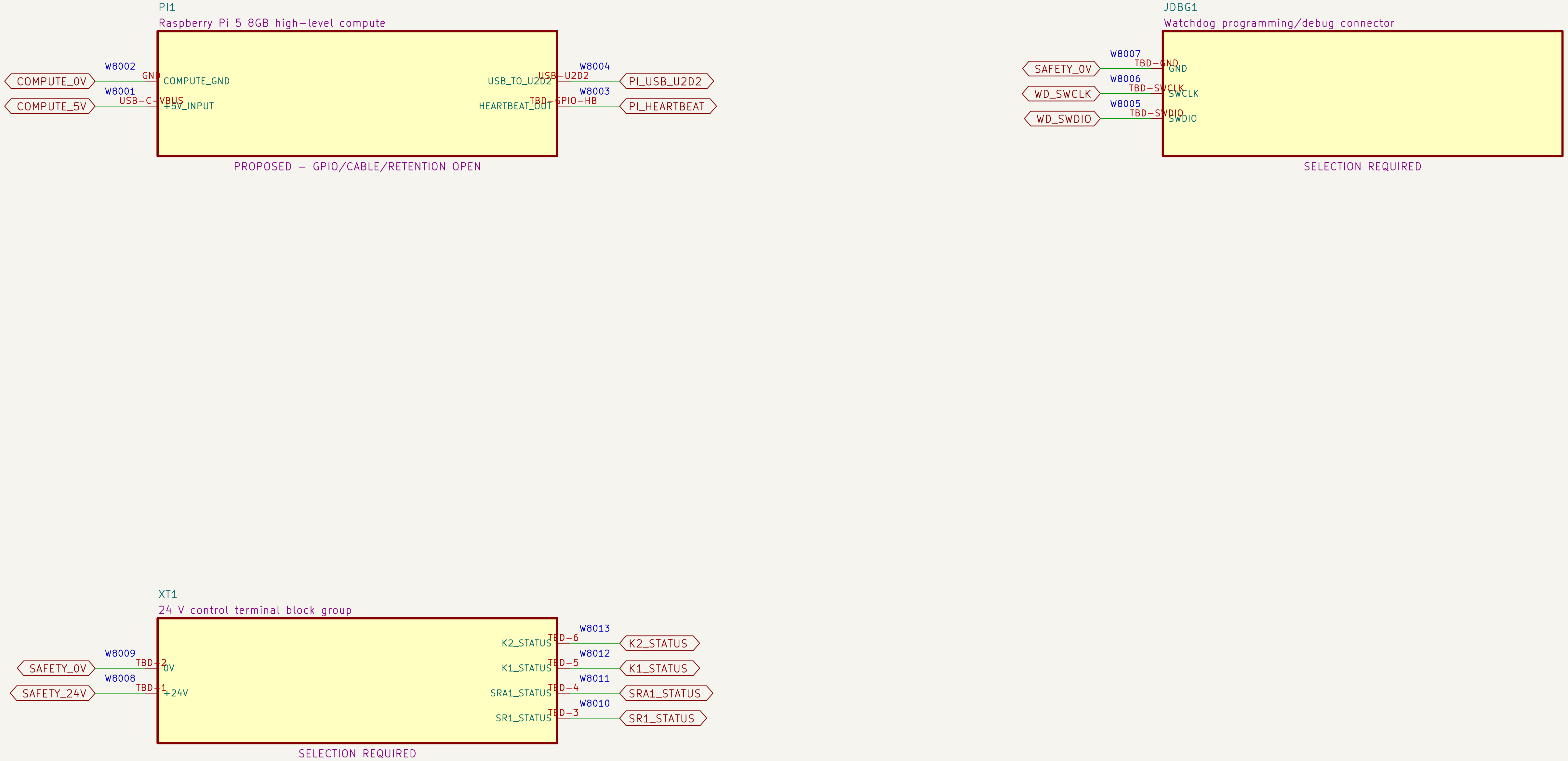
CONNECTED DESIGN CANDIDATE — QUALIFIED REVIEW REQUIRED
PRELIMINARY — NOT APPROVED FOR FABRICATION OR ENERGIZATION
Project Button
Sheet: /07 Independent watchdog power, controller and drivers/
File: 07_watchdog_control.kicad_sch

Title: PB HR-V0 Electrical V3 — 07

Size: A3	Date: 2026-08-06	**Rev: V3-P0.2**
KiCad E.D.A. 10.0.5		Id: 8/10

08 Compute, debug and control terminals

High-level compute and diagnostic wiring have no authority to bypass or restore the safety chain.



NOTE 1: Programming/debug connections must not enable outputs during operation.

NOTE 2: Terminal family, markers, conductor range, torque and enclosure layout remain selection required.

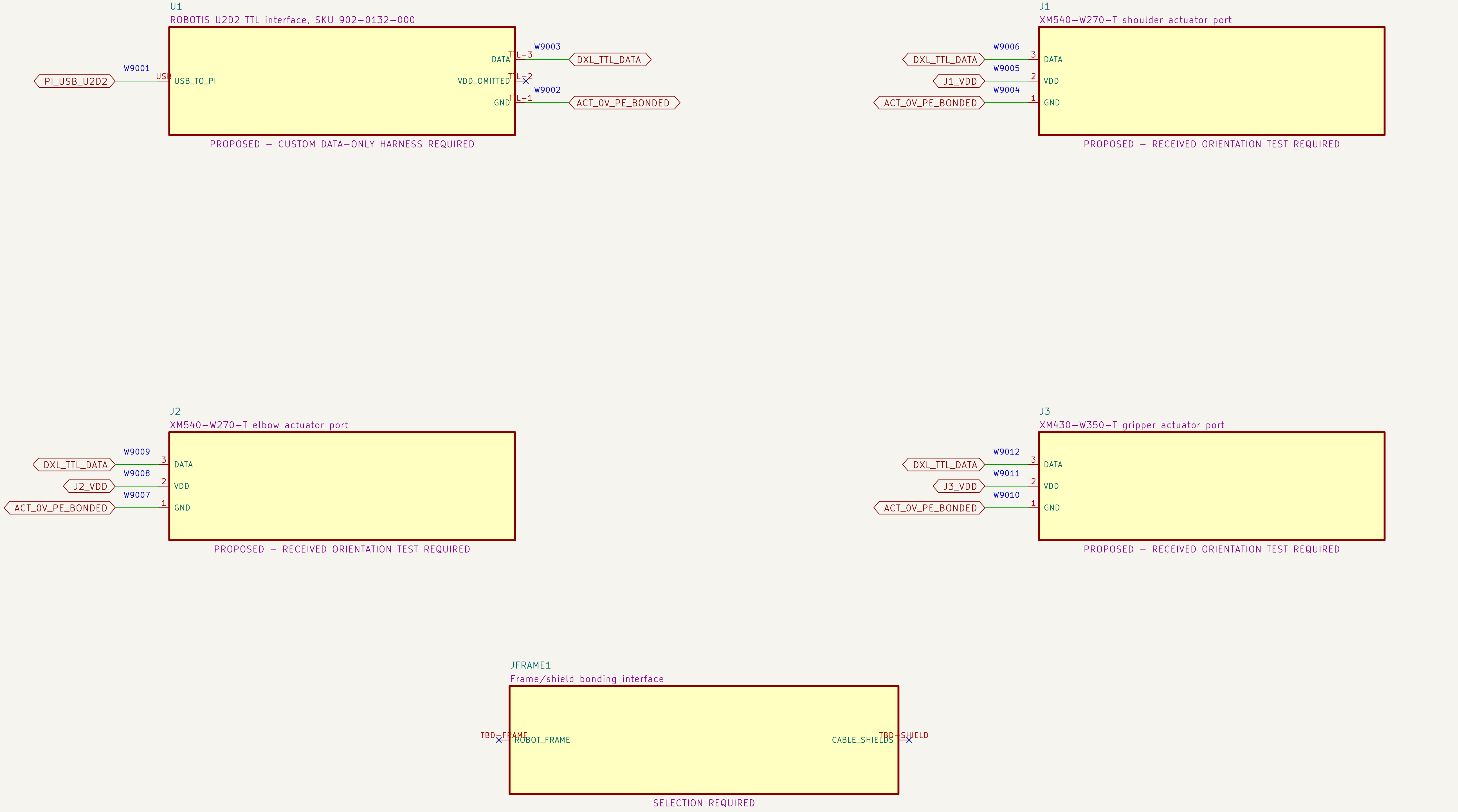
CONNECTED DESIGN CANDIDATE — QUALIFIED REVIEW REQUIRED
PRELIMINARY — NOT APPROVED FOR FABRICATION OR ENERGIZATION
Project Button
Sheet: /08 Compute, debug and control terminals/
File: 08_compute_and_control_terminals.kicad_sch

Title: PB HR-V0 Electrical V3 — 08

Size: A3	Date: 2026-08-06	**Rev: V3-P0.2**
KiCad E.D.A. 10.0.5		Id: 9/10

09 U2D2, actuator ports and bonding boundary

The U2D2 cable carries DATA and GND only; protected VDD is injected at each actuator.



NOTE 1: The V0 bus is TTL; HR-30 RS-485 remains a separate architecture.

NOTE 2: Do not bond frame or shields to 0V/PE until EMC and parallel-path review accepts the exact implementation.

CONNECTED DESIGN CANDIDATE — QUALIFIED REVIEW REQUIRED
PRELIMINARY — NOT APPROVED FOR FABRICATION OR ENERGIZATION
Project Button
Sheet: /09 U2D2, actuator ports and bonding boundary/
File: 09_actuator_interfaces.kicad_sch

Title: PB HR-V0 Electrical V3 — 09

Size: A3	Date: 2026-08-06	**Rev: V3-P0.2**
KiCad E.D.A. 10.0.5		Id: 10/10